

Excel Data Analytics Project Documentation

Project Title

Retail Sales Data Analysis Using Star Schema Data Model

Objective

The main objective of this project is to transform raw sales data into meaningful insights using data modeling and data analysis techniques.

- To analyze retail sales data using data analytics techniques.
- To build a data model by Customer, Product, Store, Sales tables.
- To create relationship using primary keys, and foreign keys.
- To perform sales analysis using Pivot Tables.
- To identify top customers, products, and store performance.
- To analyze profit, loss and discount impact on sales.
- To generate insights for business decision making.

Dataset Description

Provide details about the data used.

- **Source:** (Ecommerce Dataset its provided as part of the academic assignment for learning data analytics and data modelling concepts.)
- **File Format:** Excel (.xlsx)
- **Number of Records:**

Dimension Table	Rows
Customer Table	500
Store Table	20
Product Table	100

Fact Table	Rows
Sales Table	2000

Table Structure:

Customer Table

Column Name	Column Description	Data Type
Customer_ID	Unique ID for Each Customer	Text
Customer_Name	Name of the Customer	Text
Age	Age of the Customer	Number
Gender	Gender of the Customer	Text
City	Name of the City where the Customer is located	Text
State	Name of the State where the Customer is located	Text

Country	Name of the Country where the Customer is located	Text
Loyalty Level	Indicates the Customer membership type	Text

Store Table:

Column Name	Column Description	Data Type
Store_ID	Unique Store Identifier	Text
Store_Name	Name of the Store	Text
Region	Represent the geographical region where Store is located	Text
City	Name of the City where the Store is located	Text
Store_Type	Indicates the which type of Store	Text

Product Table:

Column Name	Column Description	Data Type
Product_ID	Unique Product Identifier	Text
Product_Name	Name of the Product	Text
Category	Product Category	Text
Sub_Category	Product Sub_Category	Text
Brand	Name of the Product Brand	Text
Cost	Represent cost of goods sold for the Product	Currency
Stock	Represent the quantity of products available in inventory	Number

Sales Table

Column Name	Column Description	Data Type
Sales_ID	Unique ID for each Sales	Text
Order_Date	Date when the order was placed	Date
Customer_ID	Unique ID for Each Customer	Text
Product_ID	Unique Product Identifier	Text
Store_ID	Unique Store Identifier	Text
Quantity	Number of units sold	Number
Unit_Price	Price per Single Unit	Currency
Discount	Discount percentage applied	Percentage
Payment_Type	Mode of Payment	Text
Total Amount	Total Revenue after discount	Currency
Gross Profit	Profit after subtracting cost	Currency
Profit/Loss	Indicate whether transaction is profit or loss	Text

Data Cleaning & Exploration

Explain preprocessing steps.

- Remove inconsistencies in the data.
- Remove degree suffixes from text.
- Trim extra spaces.
- Standardize text formatting.
- Split Full Name into First Name and Last Name.
- Change or Correct data types.
- Missing values in the **Cost column** were filled based on available data and business logic.
- Missing values in **Quantity** and **Unit Price** were detected and imputed.
- A **Gross Profit** column was created by subtracting the Cost from Total Sales for each transaction.
- A **Profit** and **Loss** column was derived from the Gross Profit column Whether the transaction resulted in a Profit or Loss.
- An **Age Group** column was generated by categorizing customer ages into predefined ranges.
- A **Discount Range** column was created by grouping discount percentages into defined brackets.
- **Month** and **Weekday** columns were derived from Order Date.

Data Formatting

- **Total Sales** values were visually emphasized in the data panel to highlight significant transactions.
- **Gross Profit** rows **greater than zero** were highlighted to identify profitable sales.
- **Profit/Loss** Column showing where the **Loss** entries are highlighted for users to quickly investigate the reasons behind the loss.

Data Analysis Techniques

List Excel tools and functions used:

- **Data Cleaning (Power Query)**

- **Pivot Table**

- **Pivot Charts**

- **Conditional Formatting**

- **Sorting & Filtering**

- **Data Imputation**

- **Functions:**

SUM ()	SUMIF ()
AVERAGE ()	AVERAGEIF ()
COUNT ()	COUNTIF ()
IF ()	MAX (), MIN ()
MODE ()	STDDEV ()
Text Functions	MONTH (), WEEKDAY ()

Key Analysis Performed

- a) **Sales Trend Analysis**

- Monthly and Yearly Sale trends analysed using pivot tables.

- b) **Regional Performance**

- Compared sales across different regions.

- c) **Category Analysis**

- Identified top-selling and low-performing products.

- d) **Discount vs Profit/Loss Analysis**

- Analyzes how discounts impact profit or loss highlighting cases where discounts lead to losses.

- e) **Customer Insights**

- Identifies top 10 customers total sales.

Visualization

Describe charts used:

- Bar Charts (for comparison)
- Column Charts (for comparison)
- Line Charts (for trends)
- Pie Charts (for distribution)
- Dashboard (interactive with slicers)

Dashboard Overview

This Dashboard provides insights into sales performance, profit and loss trends and customer behaviour.

Key Metrics:

Total Sales
Month Wise Analysis
Total Orders
Profit & Loss

Filters:

Age Group
Gender

Key Insights

Sports Category – More Profit
High Discount - Loss
Female – Major Revenue

Screenshots

Customer Table Creation

Excel Assignment 3 - Excel

Table Name: Custdb

Table Style Options: Header Row, First Column, Filter Button, Total Row, Last Column, Banded Rows, Banded Columns

Table Styles

Customer ID	Name	Age	Gender	City	State	Country	Loyalty Level
CUST1000	sheena bonilla	55	Male	Donaldton	North Carolina	USA	
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum
CUST1002	dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum
CUST1003	jeremy harris	59	Female	Thomastown	Ohio	USA	Bronze
CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA	Silver
CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze
CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA	Gold
CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA	Platinum
CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze
CUST1011	Brian Lopez	28	Female	East Krista	New York	USA	Gold
CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA	Platinum
CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA	Silver
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver
CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA	Platinum
CUST1016	John Scott	24	Female	South Christopher	Pennsylvania	USA	Bronze
CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA	Platinum
CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA	
CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA	Platinum

Product Table Creation

Excel Assignment 3 - Excel

Table Name: Proddb

Table Style Options: Header Row, First Column, Filter Button, Total Row, Last Column, Banded Rows, Banded Columns

Table Styles

Product ID	Product Name	Category	Sub Category	Brand	Cost	Stock
PROD0	Brother Laser Printer	Electronics	Computer	Hunt Inc	112.42	416
PROD1	Men's Graphic T-Shirt	Clothing	Tops	Terry-Moore	342.56	181
PROD2	Women's Casual Hoodie	Clothing	Outerwear	Sims Inc		352
P-3	Men's Running Jacket	Clothing	Outerwear	Pineda Ltd	364.67	452
PROD4	Women's Yoga Pants	Clothing	Bottoms	Gonzales-Davis	306.46	487
PROD5	Men's Leather Belt	Clothing	Accessories	Baldwin, Garza and Gonzalez	261	59
PROD6	Herbal Face Moisturizer	Beauty	Skincare	Davidson PLC	312.8	350
PROD7	LED Table Lamp	Home	Lighting	Harrell LLC	218.86	170
PROD8	Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	439.07	142
PROD9	Adjustable Dumbbell Set	Sports	Fitness	Hill PLC	495.72	323
PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	298.86	308
PROD11	HD Smart TV Stick	Electronics	Accessories	Gibson, Hobbs and Weaver	449.4	
PROD12	Hair Strengthening Shampoo	Beauty	Haircare	Rice LLC	115.82	460
PROD13	USB-C Fast Charger	Electronics	Accessories	Dawson-Perkins	242.23	453
PROD14	Decorative Wall Mirror	Home	Decor	Luna, Anderson and Bowmar	48.01	439
PROD15	4K Home Projector	Electronics	Appliances	Thomas LLC	465.34	301
P-16	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	337.42	256
PROD17	Soccer Training Cone Set	Sports	Accessories	Ford, Murphy and Wilson	144.53	367
PROD18	Women's Denim Jeans	Clothing	Bottoms	Guerra-Hendrix		343
PROD19	Knee Support Brace	Sports	Equipment	May-Reese	21.34	65

Customer Table Cleaning (Power Query)

Power Query Editor - Custdb

Table.TransformColumns(#"Capitalized Each Word4",{{"Loyalty_Level", Text.Trim, type text}})

Name	Last Name	Age	Gender	City	State	Country	Loyalty_Level
1	Bonilla	55	Male	Donaldton	North Carolina	USA	Platinum
2	Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum
3	Price	29	Female	Kylebury	Alabama	USA	Platinum
4	Harris	59	Female	Thomastown	Ohio	USA	Bronze
5	Collins	64	Female	Nicholashire	California	USA	Silver
6	Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze
7	Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum
8	Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze
9	Howard	53	Female	Maynardton	New Hampshire	USA	Gold
10	Leach	64	Male	Melissashire	Pennsylvania	USA	Platinum
11	Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze
12	Lopez	28	Female	East Krista	New York	USA	Gold
13	Lambert	55	Female	North Kathryn	Kansas	USA	Platinum
14	Hernandez	32	Female	New Becky	Montana	USA	Silver
15	Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver
16	Alvarez	49	Female	Lawsonhaven	Washington	USA	Platinum
17	Scott	24	Female	South Christopher	Pennsylvania	USA	Bronze
18	Cruz	60	Female	Brandonville	Georgia	USA	Platinum
19	Moss	43	Male	South William	Kansas	USA	Platinum
20	Torres	46	Female	Gonzalezfort	New Jersey	USA	Platinum
21	Allen	59	Male	Lake Shaneburgh	Delaware	USA	Silver
22	Martinez	36	Female	Smithburgh	Oklahoma	USA	Gold
23	Fox	33	Female	Michaelton	Hawaii	USA	Silver

9 COLUMNS, 500 ROWS

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Product Table Imputation

Excel Assignment 3 - Excel

Formula: =IF(ISBLANK([@Stock]),MEDIAN({Stock}),[@Stock])

Product ID	Product Name	Category	Sub_Category	Brand	Cost	Stock	Custom Cost	Custom Stock
PROD0	Brother Laser Printer	Electronics	Computer	Hunt Inc	112.42	416	112.42	416
PROD1	Men's Graphic T-Shirt	Clothing	Tops	Terry-Moore	342.56	181	342.56	181
PROD2	Women's Casual Hoodie	Clothing	Outerwear	Sims Inc		352	242.965	352
PROD3	Men's Running Jacket	Clothing	Outerwear	Pineda Ltd	364.67	452	364.67	452
PROD4	Women's Yoga Pants	Clothing	Bottoms	Gonzales-Davis	306.46	487	306.46	487
PROD5	Men's Leather Belt	Clothing	Accessories	Baldwin, Garza and Gonzalez	261	59	261	59
PROD6	Herbal Face Moisturizer	Beauty	Skincare	Davidson PLC	312.8	350	312.8	350
PROD7	LED Table Lamp	Home	Lighting	Harrell LLC	218.86	170	218.86	170
PROD8	Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	439.07	142	439.07	142
PROD9	Adjustable Dumbbell Set	Sports	Fitness	Hill PLC	495.72	323	495.72	323
PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	298.86	308	298.86	308
PROD11	HD Smart TV Stick	Electronics	Accessories	Gibson, Hobbs and Weaver	449.4		449.4	276
PROD12	Hair Strengthening Shampoo	Beauty	Haircare	Rice LLC	115.82	460	115.82	460
PROD13	USB-C Fast Charger	Electronics	Accessories	Dawson-Perkins	242.23	453	242.23	453
PROD14	Decorative Wall Mirror	Home	Decor	Luna, Anderson and Bowman	48.01	439	48.01	439
PROD15	4K Home Projector	Electronics	Appliances	Thomas LLC	465.34	301	465.34	301
PROD16	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	337.42	256	337.42	256
PROD17	Soccer Training Cone Set	Sports	Accessories	Ford, Murphy and Wilson	144.53	367	144.53	367
PROD18	Women's Denim Jeans	Clothing	Bottoms	Guerra-Hendrix		343	242.965	343
PROD19	Knee Support Brace	Sports	Equipment	May-Reese	21.34	65	21.34	65

Ready Accessibility: Investigate

Sales Table Imputation

Excel Assignment 3 - Excel

Table Tools

Uma Maheswari

File Home Insert Draw Page Layout Formulas Data Review View Help

Table Design Tell me what you want to do

Clipboard Font Alignment Number Conditional Formatting Styles Cell Styles Insert Delete Format Sort & Find & Filter Select Add-ins

K8

=F(IF(ISBLANK([@[Unit Price]]),AVERAGEIFS([Unit Price],[Category],[@Category],[Sub Category],[@Sub Category])),[@[Unit Price]])

	D	E	F	G	H	I	J	K	L	M	N
	Product ID	Store ID	Category	Sub Category	Quantity	Custom Quantity	Unit Price	Custom Unit Price	Discount	Custom Discot	Total Amount
2	PROD97	STORE3	Sports	Team Sports	1	1	218.04	218.04	0.09	0.09	198.4164
3	PROD74	STORE17	Electronics	Audio	3	3	104.47	104.47	0.27	0.27	228.7893
4	PROD49	STORE8	Beauty	Fragrance	4	4	914.05	914.05	0.25	0.25	2742.15
5	PROD76	STORE8	Electronics	Computer	1	1	921.51	921.51	0.16	0.16	774.0684
6	PROD59	STORE5	Beauty	Haircare	3	3	745.3	745.3	0.18	0.18	1833.438
7	PROD15	STORE7	Electronics	Appliances	3	3	147.01	147.01	0.19	0.19	357.2343
8	PROD56	STORE4	Home	Decor	2	2	785.31	785.31	0.15	0.15	1335.027
9	PROD70	STORE8	Home	Storage	4	4	342.39	342.39	0.23	0.23	1054.5612
10	PROD99	STORE5	Home	Kitchen	2	2	448.25	448.25	0.07	0.07	1250.6175
11	PROD33	STORE0	Home	Decor	2	2	873.17	873.17	0.23	0.23	1344.6818
12	PROD24	STORE14	Sports	Outdoor	1	1	278.63	278.63	0.07	0.07	259.1259
13	PROD39	STORE1	Electronics	Mobile	4	4	286.92	286.92	0.17	0.17	952.5744
14	PROD88	STORE4	Sports	Accessories	4	4	644.58	644.58	0.01	0.01	2552.5368
15	PROD34	STORE16	Electronics	Computer	2	2	316.9	316.9	0.3	0.3	443.66
16	PROD24	STORE15	Sports	Outdoor	4	4	403.47	403.47	0.29	0.29	1145.8548
17	PROD36	STORE9	Clothing	Tops	2	2	871.34	871.34	0.04	0.04	2509.4592
18	PROD65	STORE19	Sports	Fitness	2	2	451.98	451.98	0.25	0.25	677.97
19	PROD27	STORE4	Home	Decor	2	2	502.69	502.69	0.23	0.23	774.1426
20	PROD39	STORE4	Electronics	Mobile	2	2	559.23	559.23	0.25	0.25	838.845
21	PROD60	STORE13	Beauty	Fragrance	4	4	528.58	528.58	0.04	0.04	1782.4512

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Search

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Excel Formulas

VLOOKUP using Sales Table

Excel Assignment 3 - Excel

Table Tools

Uma Maheswari

File Home Insert Draw Page Layout Formulas Data Review View Help

Table Design Tell me what you want to do

Clipboard Font Alignment Number Conditional Formatting Styles Cell Styles Insert Delete Format Sort & Find & Filter Select Add-ins

M5

=VLOOKUP([@[Product_ID]],Proddb_1,3,FALSE)

	D	E	F	G	H	I	J	K	L	M	N	O	P
	Product ID	Store ID	First Name	Last Name	Age	Gender	City	State	Loyalty Lev	Category	Sub Category	Region	Store Type
2	PROD97	STORE3	Sarah	Smith	55	Male	Kathleenport	Wyoming	Gold	Sports	Team Sports	North	Outlet
3	PROD74	STORE17	Michael	Foley	63	Female	Oneilfurt	Vermont	Gold	Electronics	Audio	North	Flagship
4	PROD49	STORE8	Christopher	Sutton	49	Female	Deborahberg	Pennsylvania	Gold	Beauty	Fragrance	West	Online
5	PROD76	STORE8	John	Jordan	58	Female	Port Crystalberg	Tennessee	Platinum	Electronics	Computer	West	Online
6	PROD59	STORE5	Amanda	Powell	28	Male	South William	Alabama	Silver	Beauty	Haircare	North	Outlet
7	PROD15	STORE7	Anthony	Phillips	18	Female	North Davidport	Rhode Island	Platinum	Electronics	Appliances	South	Online
8	PROD56	STORE4	John	Scott	24	Female	South Christopher	Pennsylvania	Bronze	Home	Decor	South	Online
9	PROD70	STORE8	Patrick	Roberts	41	Female	Jaredville	Missouri	Platinum	Home	Storage	West	Online
10	PROD99	STORE5	Suzanne	Bullock	60	Female	Lake Blaketon	Alabama	Platinum	Home	Kitchen	North	Outlet
11	PROD33	STORE0	David	Pierce	45	Female	Grayland	Washington	Platinum	Home	Decor	East	Flagship
12	PROD24	STORE14	Michael	Thomas	21	Female	Woodmouth	Nevada	Silver	Sports	Outdoor	South	Outlet
13	PROD39	STORE1	Jessica	Gonzalez	34	Female	South Darrenland	Iowa	Bronze	Electronics	Mobile	North	Flagship
14	PROD88	STORE4	Wesley	Short	41	Female	Susanstad	Montana	Gold	Sports	Accessories	South	Online
15	PROD34	STORE16	Adam	Castro	51	Male	New Donnabury	Texas	Silver	Electronics	Computer	West	Flagship
16	PROD24	STORE15	Sherry	Cabrera	55	Male	Shannonview	Pennsylvania	Silver	Sports	Outdoor	North	Flagship
17	PROD36	STORE9	Lindsay	Beasley	37	Female	Wesleyhaven	Idaho	Silver	Clothing	Tops	South	Flagship
18	PROD65	STORE19	Arthur	Warren	38	Male	Erichshire	Nevada	Gold	Sports	Fitness	East	Online
19	PROD27	STORE4	Molly	Gardner	22	Female	Nicoleborough	Arkansas	Platinum	Home	Decor	South	Online
20	PROD39	STORE4	David	Patterson	43	Female	New Amyport	Nevada	Gold	Electronics	Mobile	South	Online
21	PROD60	STORE13	Maria	Hill	58	Female	Zamoraborough	Nevada	Platinum	Beauty	Fragrance	East	Flagship

Ready Accessibility: Investigate

Search

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Statistics Functions (Sum, Average, Count, Data Analysis Toolpak)

The screenshot displays an Excel spreadsheet with several data tables and statistical summaries. The 'Gender wise Sales' table shows total sales, average total sales, count, quantity purchased, profit, and loss for Male and Female. The 'Region wise Sales' table shows total sales, average total sales, orders count, and average profit for North, South, East, and West regions. The 'Loyalty Level wise Sales' table shows total sales, average sales, and count for Platinum, Gold, Bronze, and Silver loyalty levels. The 'Payment Type Analysis' table shows total sales, average sales, and count for COD, Credit Card, and PayPal payment types. A 'Total Amount' summary table provides statistical measures like Mean, Standard Error, Median, Mode, Standard Deviation, Sample Variance, Kurtosis, Maximum, Sum, and Count.

	Total Sales	Average Total Sales	Count	Quantity Purchased	Profit	Loss
Male	941940.77	1041.97	904	2190	732151.25	-18553.16
Female	1232339.80	1124.40	1096	2770	967603.55	-18607.44

	Total Sales	Average Total Sales	Orders Count	Average Profit
North	644428.25	1081.26	596	1224.78
South	634642.90	1092.33	581	1237.85
East	624595.90	1058.64	590	1193.98
West	270613.51	1161.43	233	1261.26

	Total Sales	Average Sales	Count
Platinum	673152.51	1082.24	622
Gold	518439.54	1105.41	469
Bronze	523385.63	1074.71	487
Silver	459302.89	1088.40	422

	Total Sales	Average Sales	Count
COD	702179.37	1063.91	660
Credit Card	716037.87	1110.14	645
PayPal	756063.32	1087.86	695

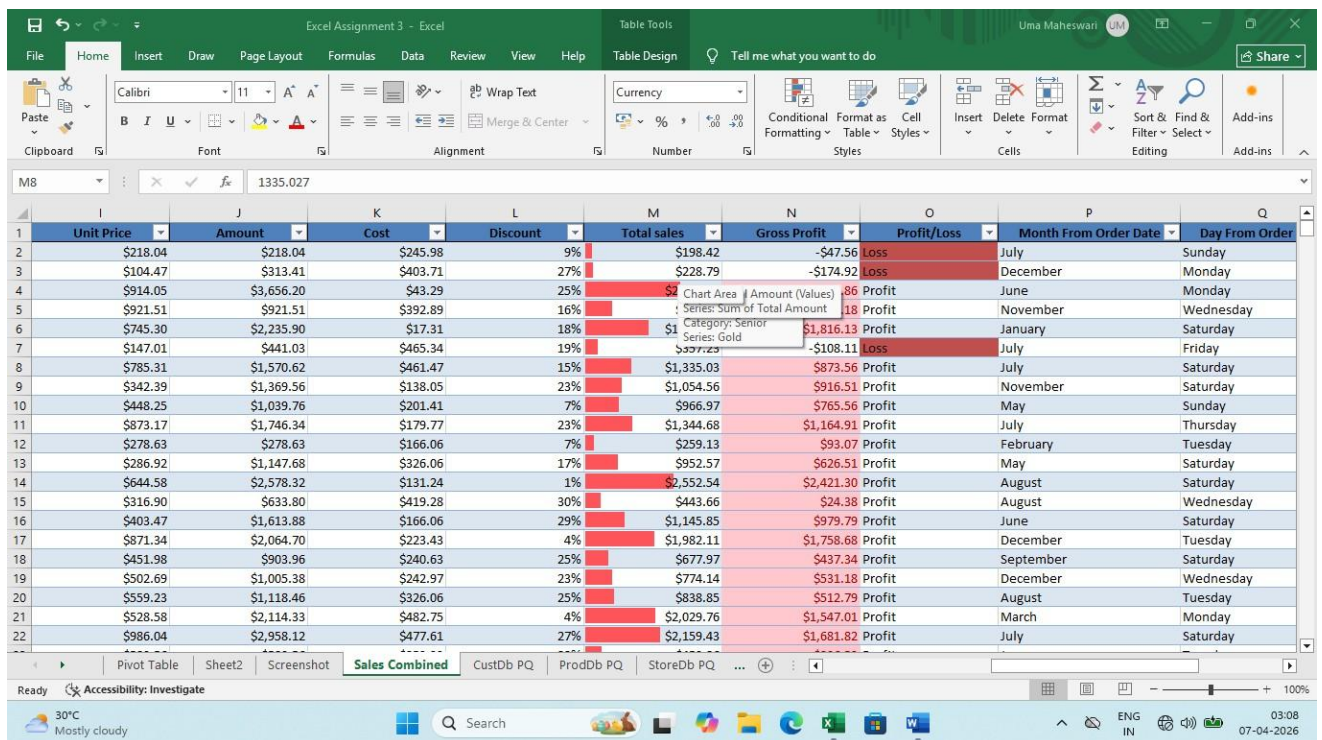
Mean	1087.140283
Standard Error	18.30700645
Median	871.40815
Mode	#N/A
Standard Deviation	818.7142176
Sample Variance	670292.9701
Kurtosis	0.053514273
Maximum	3908.2428
Sum	2174280.566
Count	2000

Data Exploartion: (New Column Created) [Gross Profit, Profit/Loss, Age group, Discount Ranged, Year, Day, Text function]

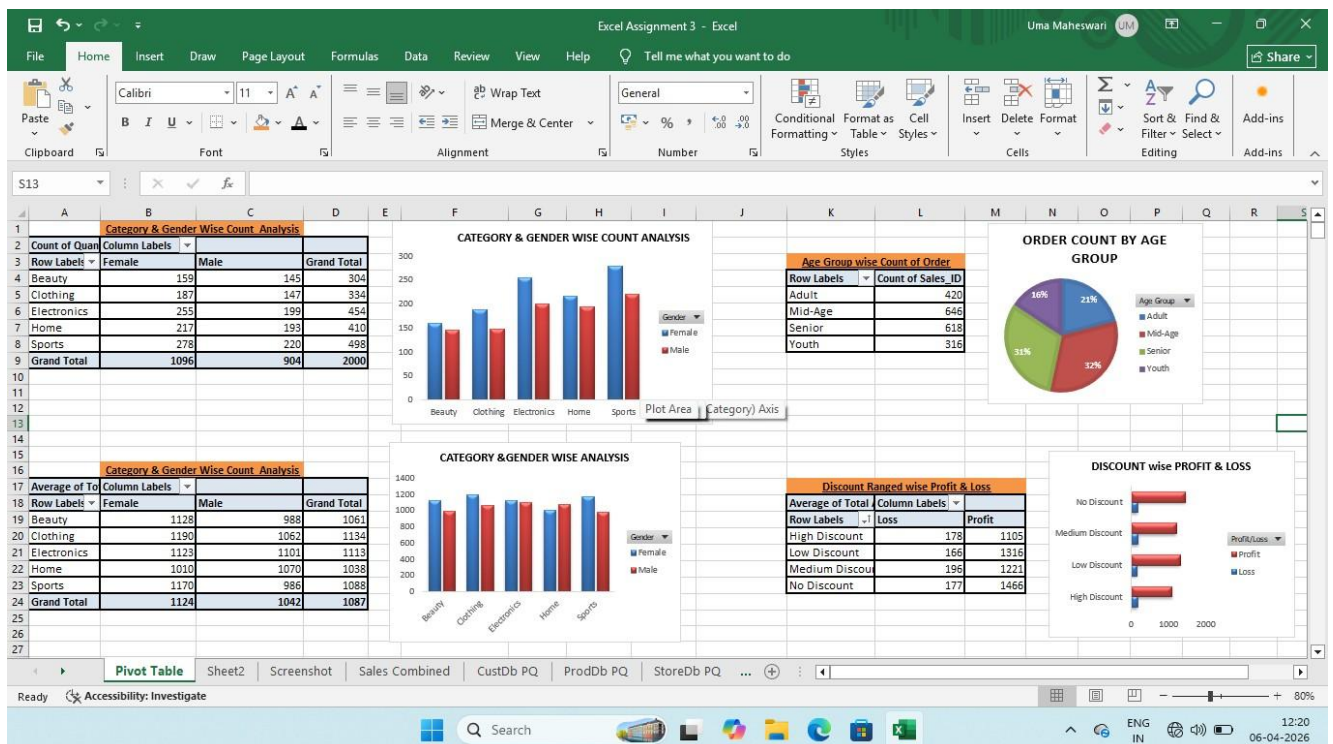
The screenshot displays an Excel spreadsheet with a detailed data table. The table includes columns for Total sales, Gross Profit, Profit/Loss, Month From Order Date, Day From Order Date, Age Group, Discount Ranged, Loyalty Level, Year, and Month. The data is filtered by a formula in cell S6: =IF([@Discount]=0,"No Discount",IF([@Discount]<=0.1,"Low Discount",IF([@Discount]>0.1,"High Discount"])). The table shows various sales transactions with their respective profits, discounts, and loyalty levels.

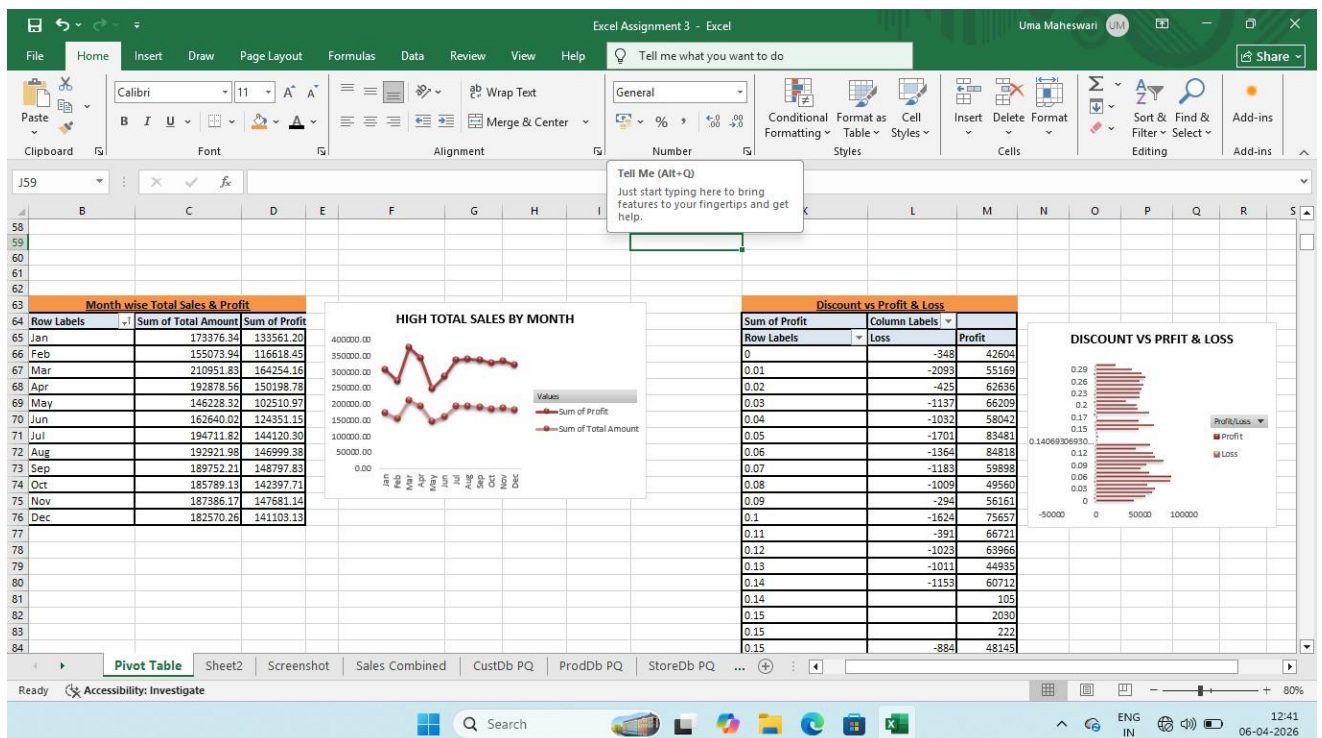
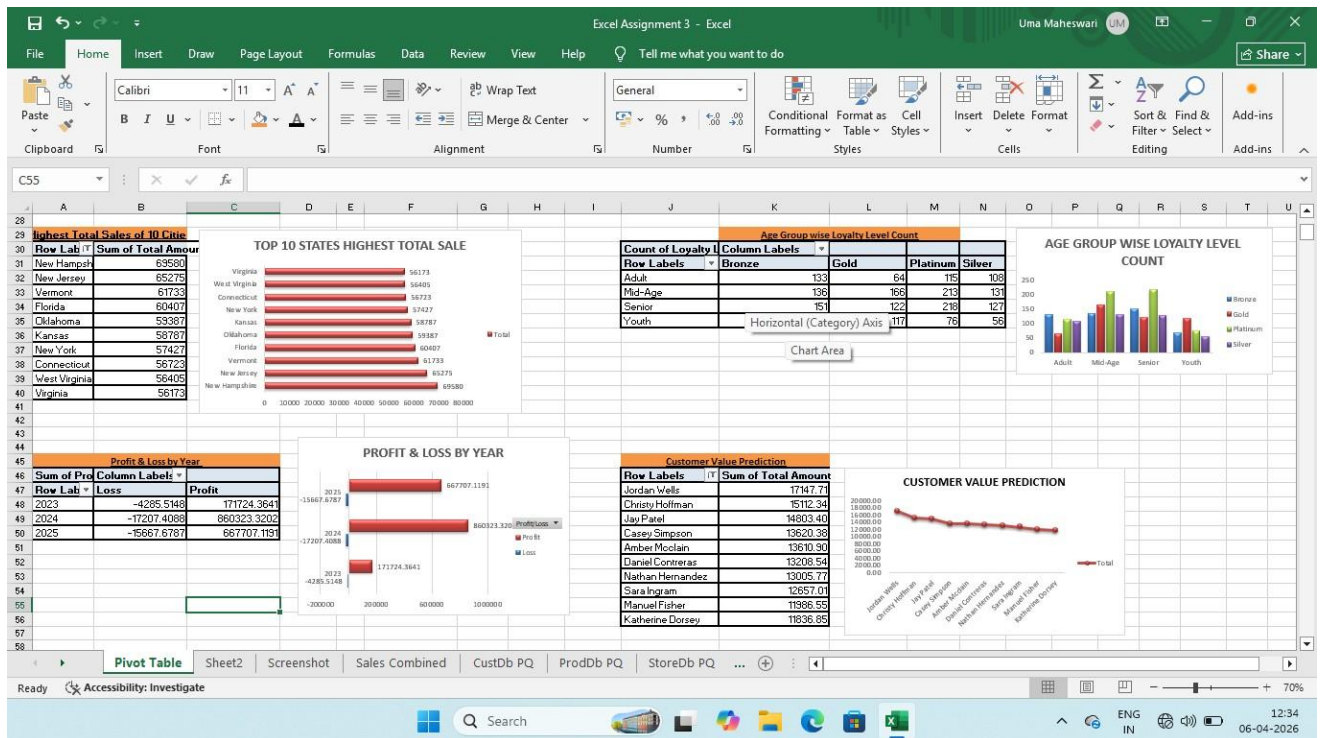
	Total sales	Gross Profit	Profit/Loss	Month From Order Date	Day From Order Date	Age Group	Discount Ranged	Loyalty Level	Year	Month
1	\$198.42	-\$47.56	Loss	July	Sunday	Senior	Low Discount	Gold	2024	7
2	\$228.79	-\$174.92	Loss	December	Monday	Senior	High Discount	Gold	2023	25
3	\$2,742.15	\$2,698.86	Profit	June	Monday	Mid-Age	High Discount	Gold	2024	3
4	\$774.07	\$381.18	Profit	November	Wednesday	Senior	Medium Discount	Platinum	2023	29
5	\$1,833.44	\$1,816.13	Profit	January	Saturday	Adult	Medium Discount	Silver	2024	6
6	\$357.23	-\$108.11	Loss	July	Friday	Youth	Medium Discount	Platinum	2024	26
7	\$1,335.03	\$873.56	Profit	July	Saturday	Youth	Medium Discount	Bronze	2025	12
8	\$1,054.56	\$916.51	Profit	November	Saturday	Mid-Age	High Discount	Platinum	2023	18
9	\$966.97	\$765.56	Profit	May	Sunday	Senior	Low Discount	Platinum	2025	4
10	\$1,344.68	\$1,164.91	Profit	July	Thursday	Mid-Age	High Discount	Platinum	2025	31
11	\$259.13	\$93.07	Profit	February	Tuesday	Youth	Low Discount	Silver	2024	13
12	\$952.57	\$626.51	Profit	May	Saturday	Adult	Medium Discount	Bronze	2025	24
13	\$2,552.54	\$2,421.30	Profit	August	Saturday	Mid-Age	Low Discount	Gold	2024	24
14	\$443.66	\$24.38	Profit	August	Wednesday	Senior	High Discount	Silver	2025	6
15	\$1,145.85	\$979.79	Profit	June	Saturday	Senior	High Discount	Silver	2025	21
16	\$1,982.11	\$1,758.68	Profit	December	Tuesday	Mid-Age	Low Discount	Silver	2024	10
17	\$677.97	\$437.34	Profit	September	Saturday	Mid-Age	High Discount	Gold	2025	13
18	\$774.14	\$531.18	Profit	December	Wednesday	Youth	High Discount	Platinum	2024	11
19	\$838.85	\$512.79	Profit	August	Tuesday	Mid-Age	High Discount	Gold	2024	13
20	\$2,029.76	\$1,547.01	Profit	March	Monday	Senior	Low Discount	Platinum	2024	18
21	\$2,159.43	\$1,681.82	Profit	July	Saturday	Senior	High Discount	Bronze	2025	5

Conditional Format

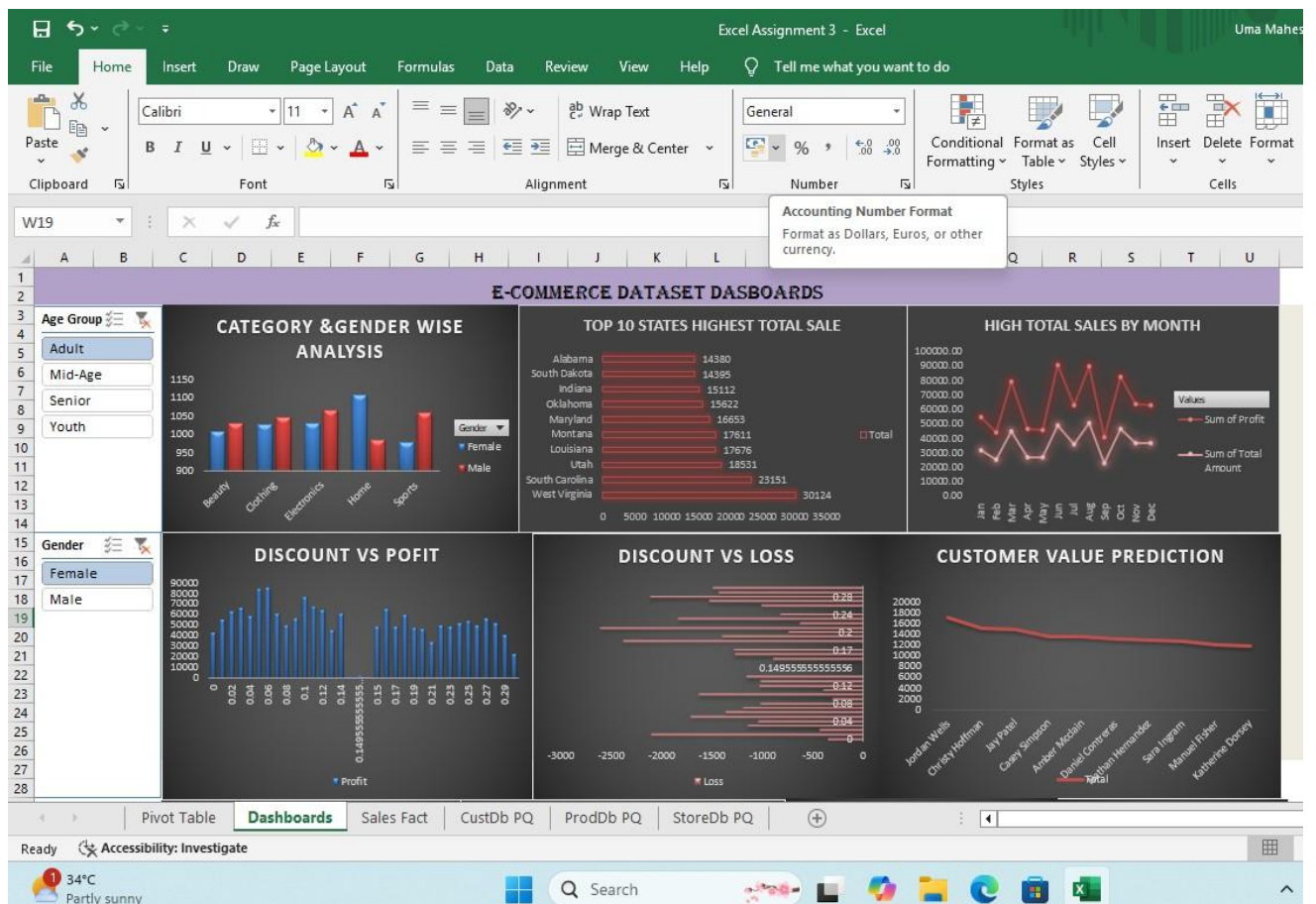


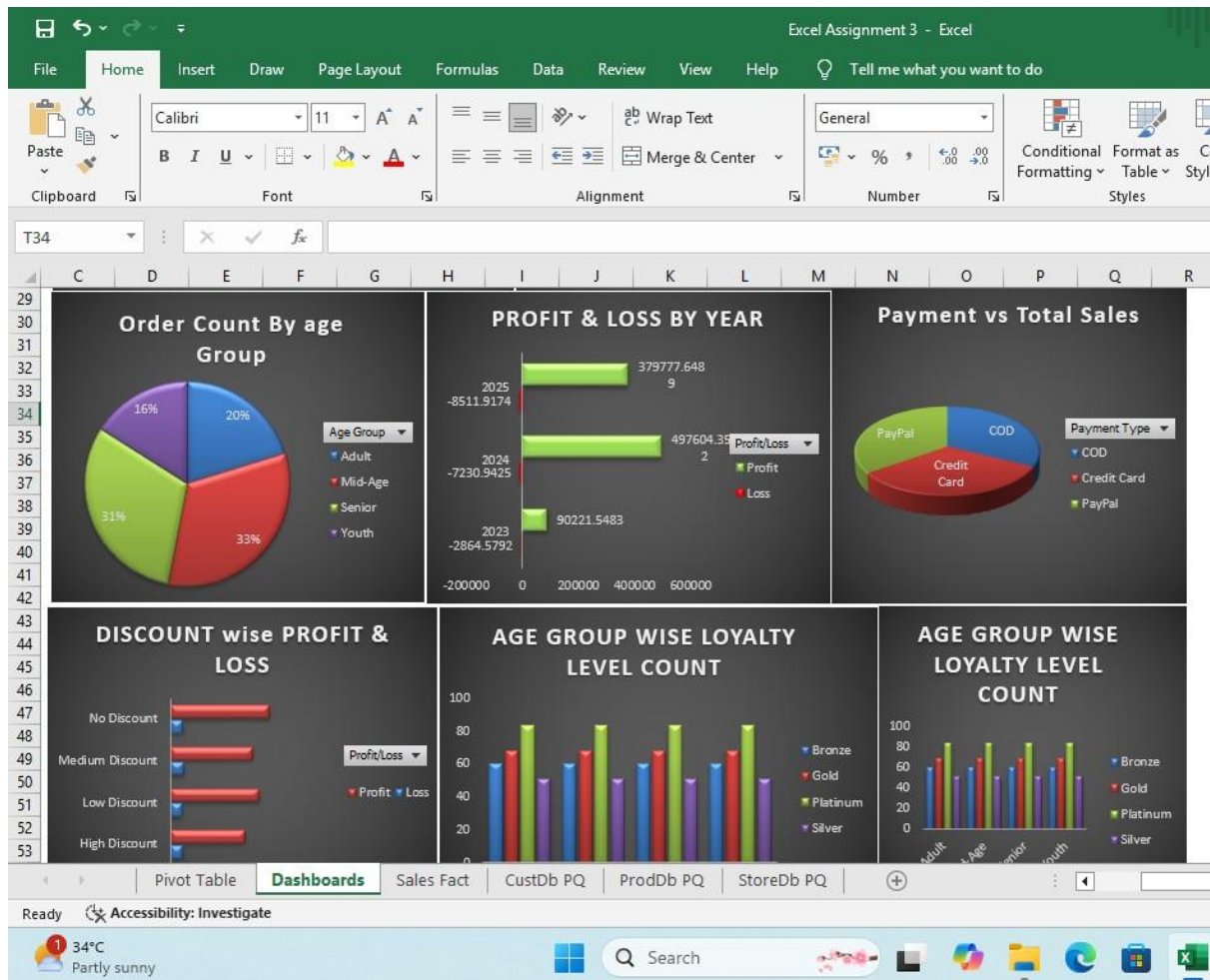
Pivot Tables





Dashboard Images





Insights:

Descriptive Analysis:

- Female Customers have a higher order count and purchase quantity, which contributes to their total Sales.
- The North region shows strong Customers demand, with higher order count and quantity leading to the highest total sale
- PayPal is the most preferred payment method used by a majority of customers.
- Customers in the middle age group contributes the highest purchases compared to other age group.

Diagnostic Analysis:

- The West region has lower sales due to significantly lower order quantity (593) compared to other regions (1400- 1550).
- Losses occur regardless of discount levels as the cost price exceeds the selling price indicating a pricing issue.

Predictive Analysis

- Female customers are expected to contribute more to future sales due to their higher order count and purchase quantity.
- The middle age group is likely to remain the key contributor to total sales.

Prescriptive Analysis:

- Maintain a selling price above cost price and avoid excessive discounts to ensure profitability.
- Focus marketing efforts on the West region to increase customer engagement.

Conclusion

The analysis highlights key sales drivers and problem areas. By improving pricing strategies, controlling discounts and focusing on high performing segments the business can enhance both sales and profitability.
